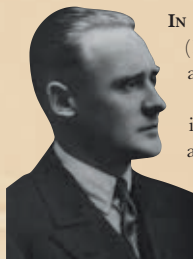


Dewoitine D.520 fighter, had only just begun to come into service when the German *Blitzkrieg* struck in 1940.

Soviet innovators

Hitler had made it clear throughout his political career that the “Jewish Bolsheviks” of the Soviet Union were intended for destruction. His rise to power brought a definitive end to the links between the Soviet and German aviation establishments that had served both so well in the 1920s and early 30s. Even in that decade, the Soviet Union had put considerable resources behind the creation of its air force and nascent air industry. Many of the leading talents in Russian aviation, including Igor Sikorsky, Alexander Seversky, and Alexander Kartveli, had gone into exile after the revolution, contributing instead to the progress of aviation in the United States – Kartveli, for example, designed the Republic P-47 Thunderbolt, one of the great American aircraft of World War II. But a tradition of aircraft design and aerodynamic research was firmly implanted. Andrei Tupolev, who as an engineering student in prewar Russia had been arrested by the czarist police for his revolutionary activities, emerged in the 1920s as the prime mover in Soviet aviation. Other talented individuals who rose through the Soviet system included

REGINALD MITCHELL



IN 1933 REGINALD MITCHELL (1895–1937), chief designer at the Supermarine aircraft company, took a vacation in Europe to convalesce after undergoing surgery for cancer. A conversation with some German aviators convinced him that war was on its way, and from that moment

he devoted himself single-mindedly, and against medical advice, to the creation of the fighter that would be called the Spitfire.

Born in Stoke-on-Trent, in England's industrial Midlands, Mitchell was an apprentice railroad engineer before joining the Supermarine Aviation Works in Southampton in

1917. Within two years he had become the company's chief designer, and when Supermarine was taken over by Vickers in 1928, it was mainly because they wanted Mitchell's services.

He made his reputation designing seaplanes for the Schneider Trophy; his 1925 S.4 gave the world its first view of the kind of fast, streamlined monoplane that was to be his specialty. Known for his attention to detail, he went on to design the Schneider-winning S.5 and S.6 – one version of which became the first aircraft to top 400mph (640kph) in 1931. Exhausted by his work on the Spitfire, Mitchell died at the age of 42, shortly before it went into production, but he was already sure of the aircraft's success. His only regret was the name “Spitfire,” dreamed up by Vickers.

SUPER SPITFIRE

Mitchell's design for the Spitfire was revolutionary in its use of an elliptical wing, designed for maximum aerodynamic efficiency. The Spitfire evolved through many versions, late models being substantially different from the 1937 prototype (right).



Sergei Ilyushin (eventually assigned exclusively to the development of long-range bombers), Nikolai Polikarpov, Alexander Yakovlev, and Syemyen Lavochkin, creator of the Lavochkin La-5 fighter.

Through the 1930s, these designers had to cope with working under the increasingly paranoid rule of Soviet dictator Joseph Stalin. As early as 1929, Polikarpov was arrested for “sabotage” when his department's development of a new fighter fell behind schedule. Along with his entire design team, he was consigned to the prison section of a state aviation factory (the use of prison labor was an important element in the Soviet economy), where they designed the Polikarpov I-5 fighter, earning their release in 1933. Tupolev himself was one of the thousands of prominent individuals “purged” by Stalin in the second half of the 1930s, spending six years working on aircraft design in one of the “special camps” of the Gulag. He was only released in 1943.

Yet within this bizarre system some excellent innovative aircraft design was achieved. Polikarpov, for example, was responsible for the I-16 which, along with the Messerschmitt Bf 109, was one of the very first single-seat, low-wing monoplane fighters. And Stalin's ruthless drive to industrialize the Soviet Union in the 1930s, although carried out at the expense of great

human suffering, did create the basis for an effective mass-production aircraft industry. But the impact of the Stalinist purges of the late 1930s on the Soviet air force was devastating. About three-quarters of senior officers were either executed or sent to the Gulag labor camps. The effects of this blow were still being felt when Germany invaded the Soviet Union in 1941.

New generation of fighters

In all the major air forces, the 1930s saw the same progress from biplane fighters – the sort of airplanes seen attacking King Kong on the top of the Empire State Building in the famous 1933 movie – to sleek cantilever-wing monoplanes such as the Spitfire, Bf 109, or the Fiat G.50 Freccia. The new generation of fighters consisted mostly of metal airplanes (although the Hurricane, one of the most successful, had a fabric-covered fuselage supported in part by wooden strips). They mounted a powerful engine in a lightweight frame and were designed with a scrupulous eye to reducing drag, not only abolishing the old biplane struts and wires but also having a retractable undercarriage and guns that were built into the wings or fuselage. From a traditional pilot's point of view, their most controversial aspect was an enclosed cockpit and an implied dependence on instrumentation. When

